Experiment on Verification of Kirchhoff's Current Law (KCL)

AIM

To experimentally verify Kirchhoff's Current Law

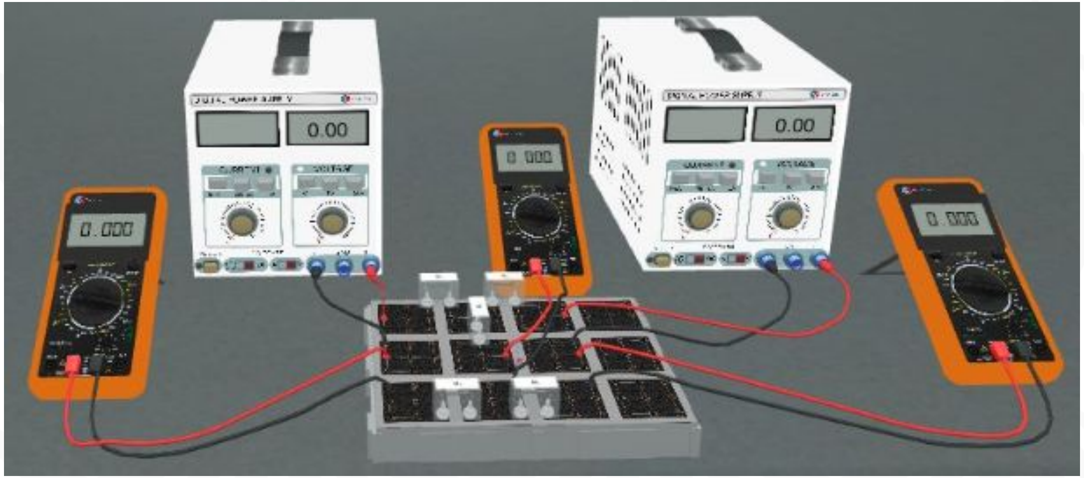
OBJECTIVES

At the end of this lesson, you should be able to:

- State Kirchhoff's Current Law (KCL) for electric circuits.
- Apply Kirchhoff's current law at node (or junction) points in an electric circuit.

APPARATUS

- Breadboard.
- Two DC power supplies.
- Five variable resistors.
- Three ammeters.
- Connection cables.



THEORY

Analysing electric circuits depends on two basic laws that were developed by the Russian scientist Justaff Kirchhoff in 1845.

- Kirchhoff's First Law (or current law) KCL
- Kirchhoff's Second Law (or Voltage law) KVL

Kirchhoff's Current Law (KCL)

KCL is based on the principle of conservation of charge.

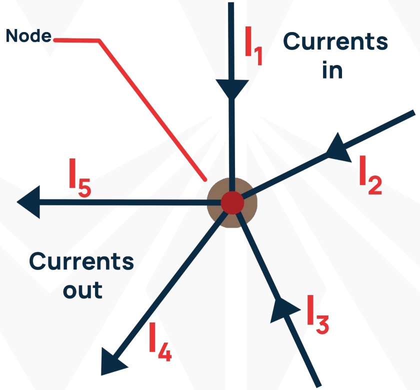
It states that:

The sum of currents that enter the junction equals the sum of currents that leave the junction.

Or

The sum of all currents entering and leaving a junction point in a circuit is zero, with currents entering the junction considered positive and those leaving the junction considered negative.

$$\sum I_{in} = \sum I_{out} \qquad \sum I = 0$$



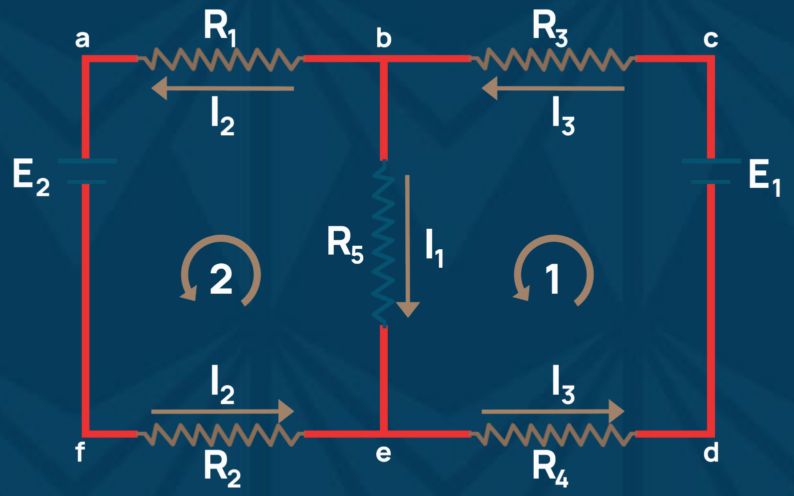
$$I_1 + I_2 + I_3 + (-I_4 - I_5) = 0$$

THEORY

At node b in the figure,

$I_1 + I_2 = I_3$ (1)

$-I_1 - I_2 + I_3 = 0$ (Current Entering is +ve)



MID-LESSON QUESTIONS

Q1: Explain Kirchhoff's Current Law (KCL) in your own words.

Q2: You have a circuit with three branches connected at a junction. If you measure a current of 2 amperes entering the junction from one branch, how much total current should be leaving the junction?

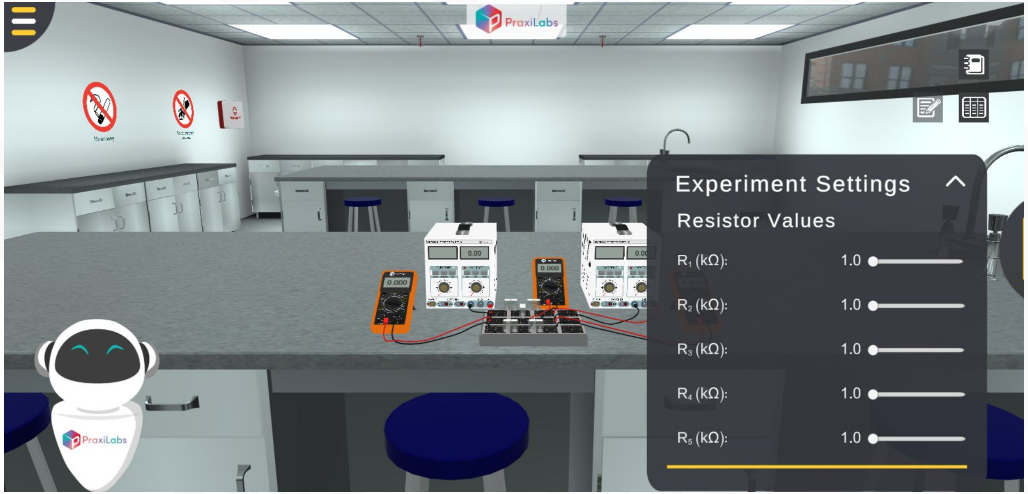
MID-LESSON ANSWERS

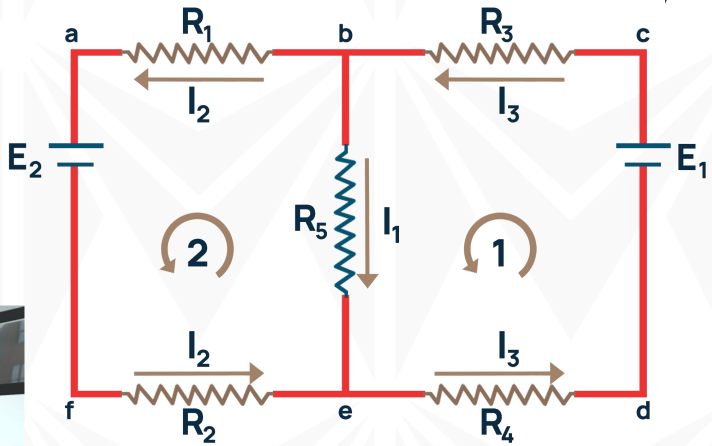
A1: Kirchhoff's Current Law (KCL) states that the total current entering a junction or node in an electrical circuit must be equal to the total current leaving that junction or node. In other words, the sum of currents at any point in a circuit is conserved, signifying the principle of charge conservation.

A2: According to Kirchhoff's Current Law (KCL), if 2 amperes of current are entering the junction, the total current leaving the junction must also be 2 amperes. KCL ensures that the conservation of charge is maintained at the junction, so the sum of currents entering and leaving must be the same.

PROCEDURE

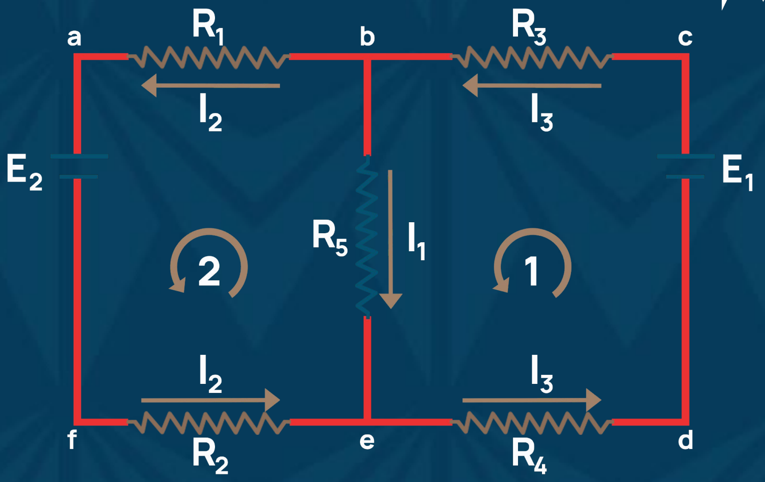
1. Select the values of R_1 through R_5 and connect the circuit as shown.

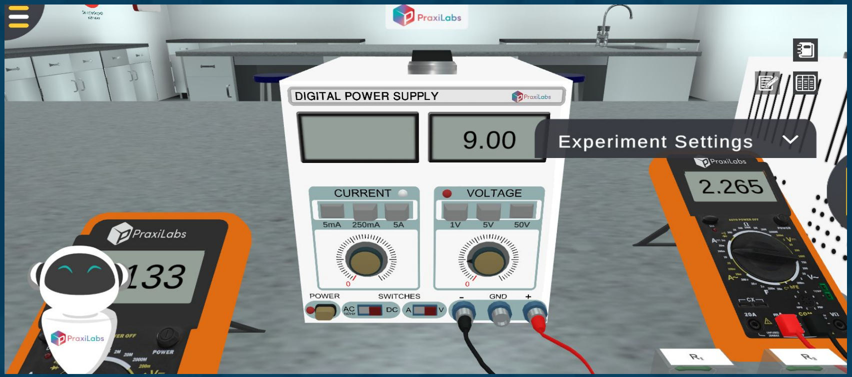


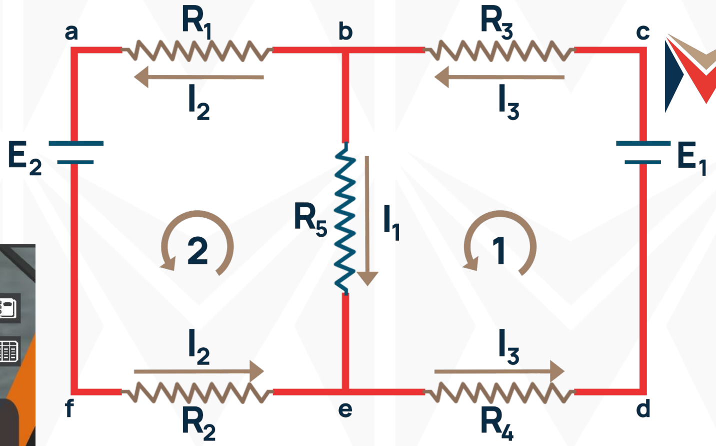


PROCEDURE

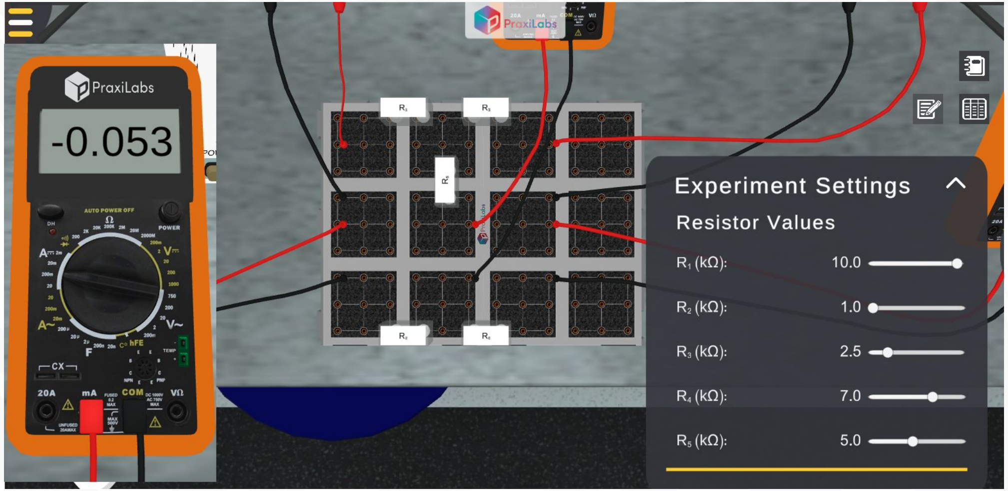
2. Select the maximum output voltage of both DC power supplies to be 50 V by pressing the 50 V button above the voltage dial.
3. Adjust the emfs of both supplies to 9 V using the voltage dial knob.







PROCEDURE



4. Click the Record button to record the readings of the three ammeters.
5. Increase the EMF E2 at regular intervals (3 V) and repeat step 4

PROCEDURE

6. Save the Excel file you downloaded

R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	R_4 (kΩ)	R_5 (kΩ)	E_2 (V)	E_1 (V)	I_1 (mA)	I_2 (mA)	I_3 (mA)	$I_{3\text{-computed}}$ (mA)
10	1	2.5	7	5	9	9	0.895	-0.416	0.479	
10	1	2.5	7	5	9	12	1.055	-0.344	0.711	
10	1	2.5	7	5	9	15	1.215	-0.271	0.944	

Data Analysis

R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	R_4 (kΩ)	R_5 (kΩ)	E_2 (V)	E_1 (V)	I_1 (mA)	I_2 (mA)	I_3 (mA)	$I_{3\text{-computed}}$ (mA)
10	1	2.5	7	5	9	9	0.895	-0.416	0.479	
10	1	2.5	7	5	9	12	1.055	-0.344	0.711	
10	1	2.5	7	5	9	15	1.215	-0.271	0.944	

1. Using the KVL equation $I_1+I_2 = I_3$ (1)

1. Calculate the $I_{3\,compound}$ using I_1 and I_2 as shown in the table

1. Compare between the values calculated and the experimental values of I_3 and then comment on your results.

WALK THROUGH VIDEO

FURTHER READING

Bueche F. and Hecht E. Schaum's Easy Outlines - College Physics Crash Course. 2000 McGraw-Hill, USA.

Hernández-Hall, C. A., & Wilson, J. D. (2014). Physics Laboratory Experiments. Cengage Learning.

Sang, D., Hewlett, C., Jones, G., Field, S., & Styles, D. (2020). Cambridge International AS & A Level Physics Practical Workbook. Cambridge University Press.

Serway, R.A. and Jewett. J.W. (2014). Physics for Scientists and Engineers with Modern Physics, Ninth Edition. Physical Sciences: Mary Finch.

FURTHER READING

Shukla, R. K., & Srivastava, A. (2006). Practical Physics. New Age International (P) Limited, Publishers.

Young, H. D. (2012). Sears & Zemansky's college physics. 9th ed. Pearson Education, Inc.

http://hyperphysics.phy-astr.gsu.edu

http://www.physicstutorials.org

Thank You